

COMP6237 Data Mining

Lecture 7: Decision Trees

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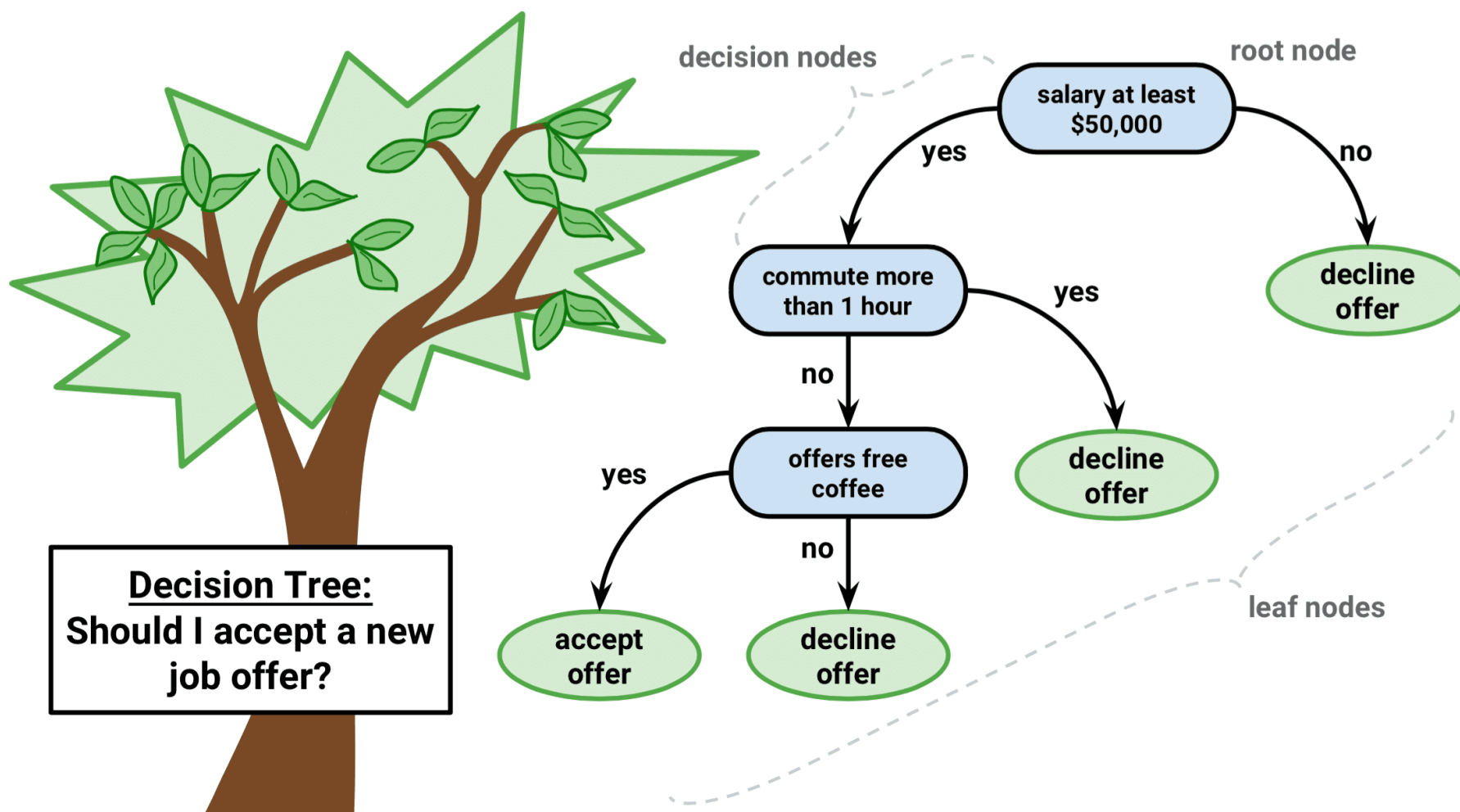
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University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Decision Trees – Problem



Source: <http://dataaspirant.com/2017/01/30/how-decision-tree-algorithm-works/>

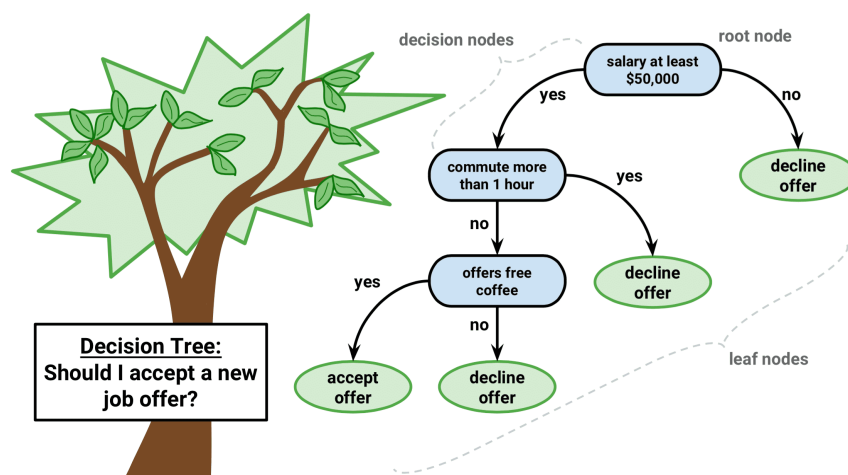
Decision Trees – Problem

We encode an offer as features and a label:

- Salary (in £k): `salary_k`
- Commute time (hours): `commute_h`
- Free coffee? (0/1): `free_coffee`
- Label: `accept` (1 = accept, 0 = decline)

	salary_k	commute_h	free_coffee	accept
0	45	0.5	1	0
1	48	1.2	1	0
2	49	0.8	0	0
3	52	0.7	0	0
4	55	0.9	1	1
5	57	1.3	1	0
6	60	0.6	1	1

Problem: Build up a decision tree to fit the data (learn the decision or grouping rules)



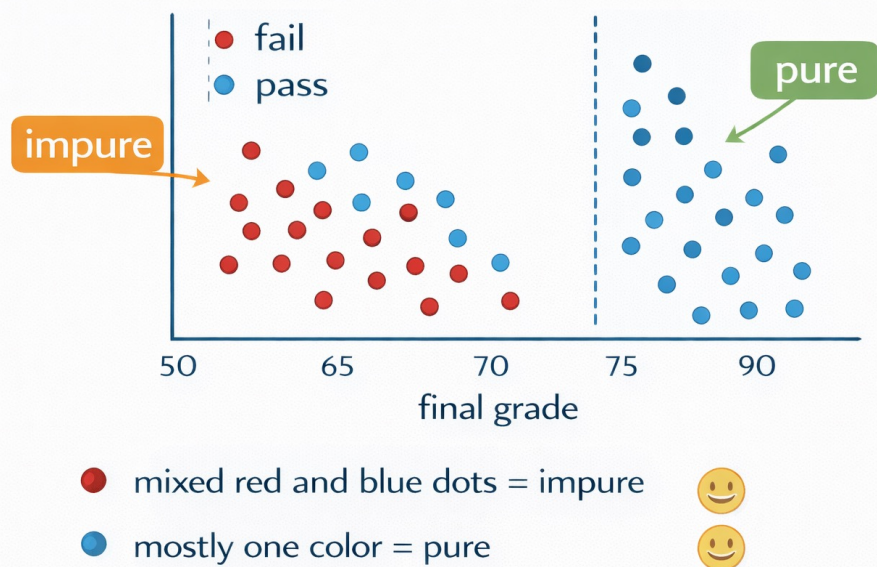
Decision Trees – Approach: CART

CART: Classification and Regression Trees

One of the most famous decision tree algorithms is CART.
Published in 1984 (!) — even older than most of us in here!

It builds decision trees by repeatedly splitting the data. And every time it considers a split, it uses something called an impurity/Gini measure.

What is “purity” vs “impurity”?

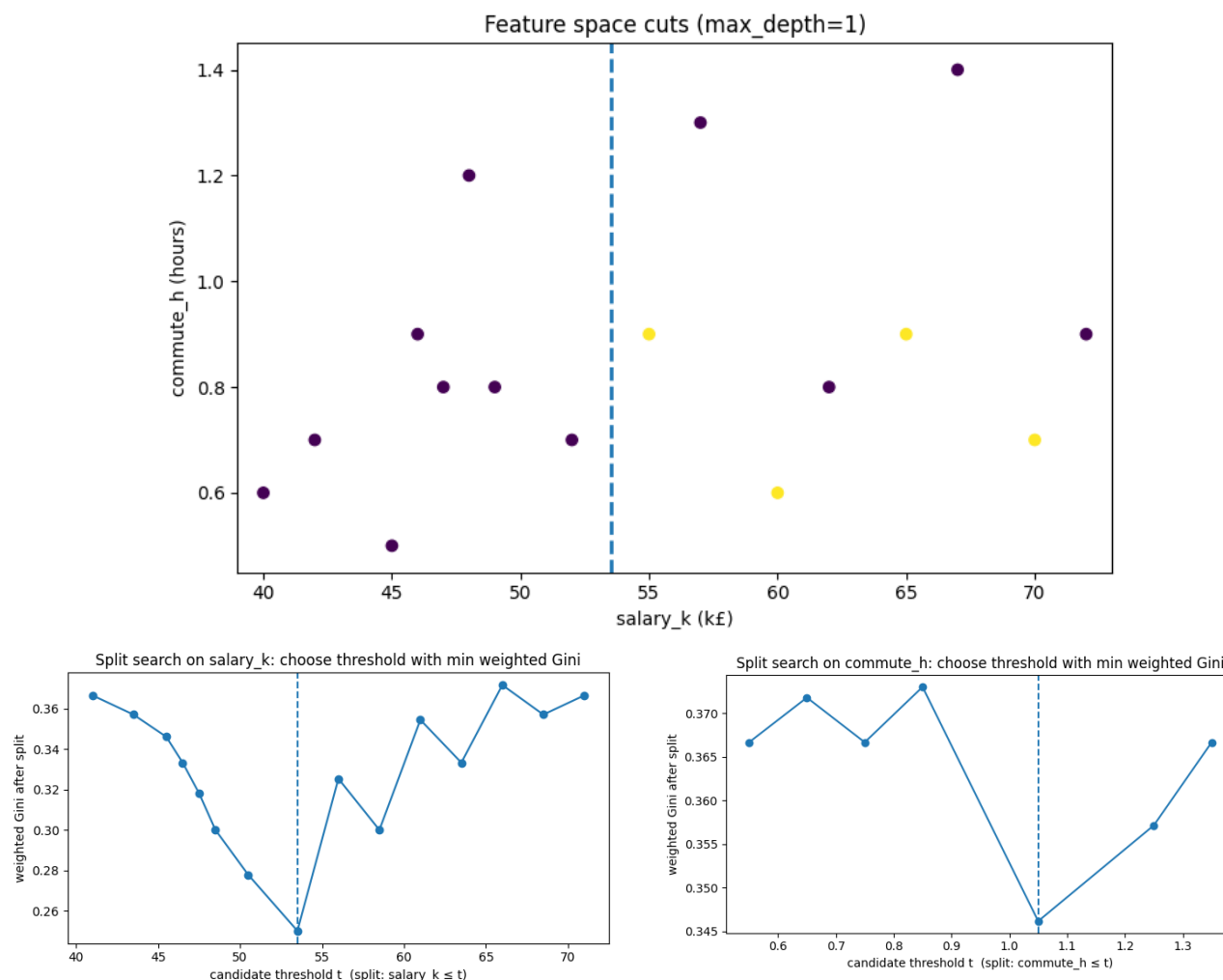


Build a decision tree as repeatedly asking: “What split makes the groups purer and purer?”

Decision Trees – Approach: CART

Key idea: Find the best split with the lowest impurity/Gini value

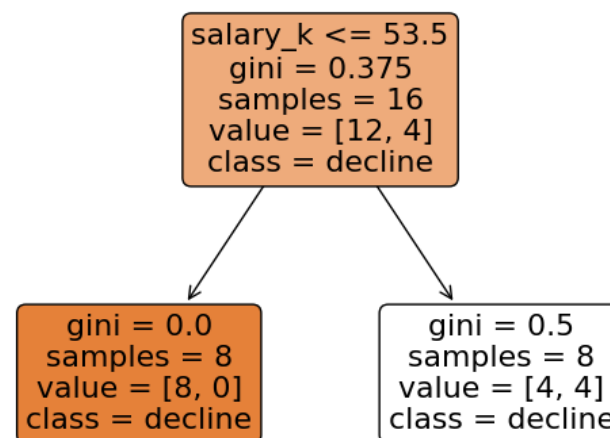
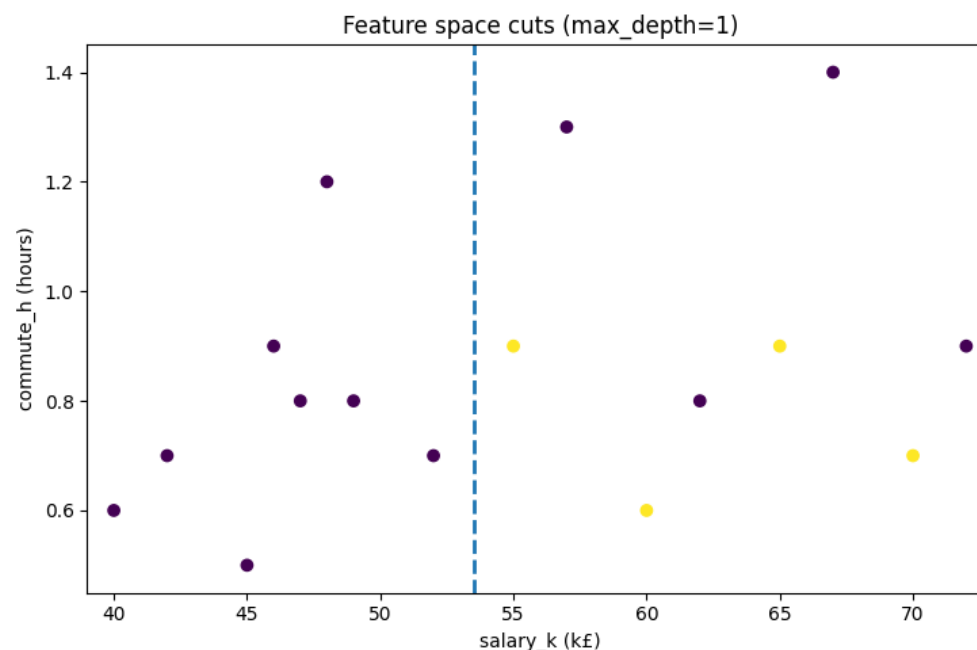
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2	49	0.8	0
3	52	0.7	0
4	55	0.9	1
5	57	1.3	0
6	60	0.6	1
7	62	0.8	0
8	65	0.9	1
9	67	1.4	0
10	70	0.7	1
11	72	0.9	0
12	40	0.6	0
13	42	0.7	0
14	46	0.9	0
15	47	0.8	0



Decision Trees – Approach: CART

Key idea: Find the best split with the lowest impurity/Gini value

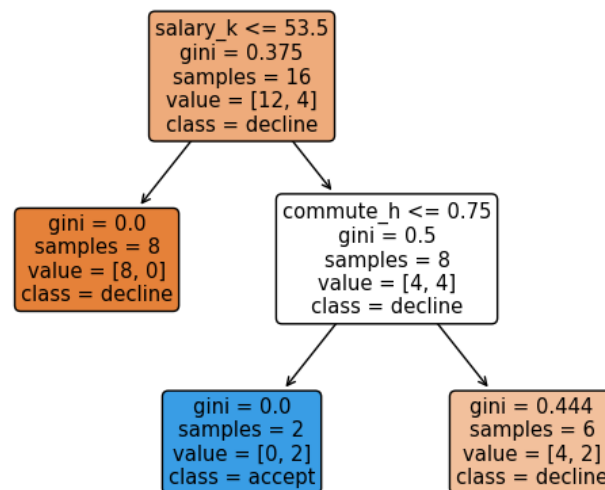
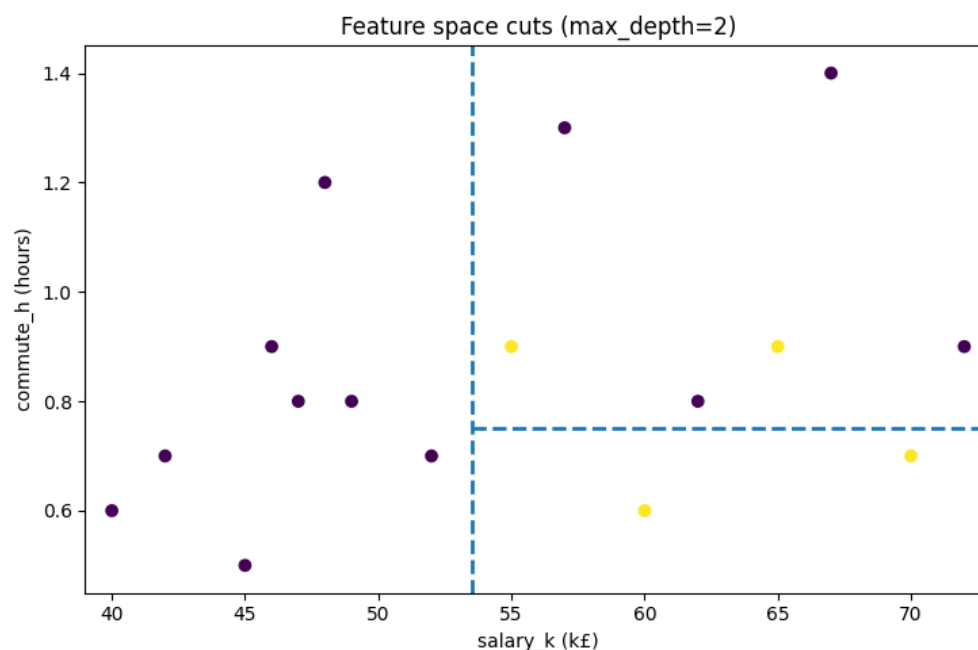
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Decision Trees – Approach: CART

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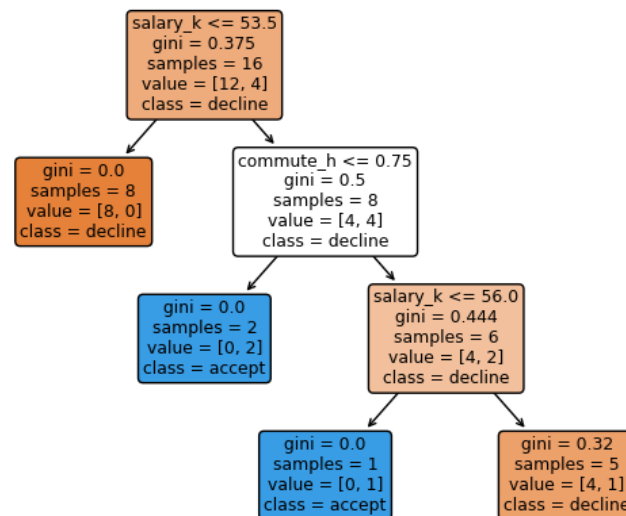
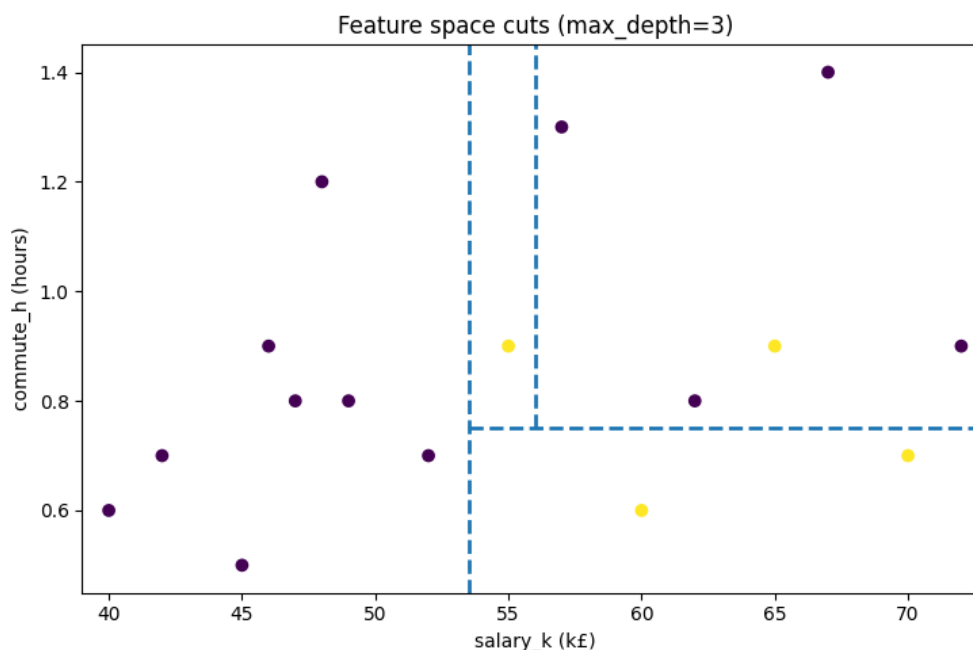
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Decision Trees – Approach: CART

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Decision Trees – Approach: CART

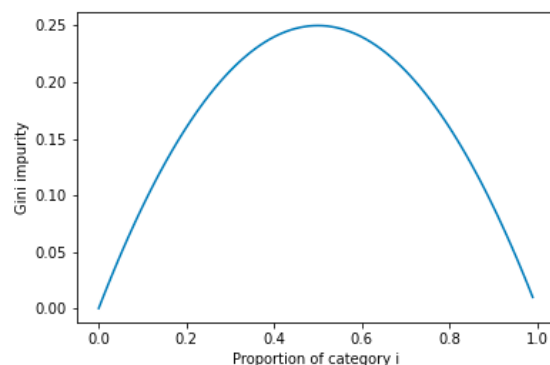
- **Gini Index** for a given node t :

The smaller
the purer

$$GINI(t) = 1 - \sum_j^{n_c} [p(j|t)]^2$$

(where: n_c is the number of classes, and $p(j | t)$ is the relative frequency of class j at node t).

- Maximum: $(1 - 1/n_c)$ when records are equally distributed among all classes, implying least interesting information
- Minimum: (0.0) when all records belong to one single class, implying most interesting information



It reaches the minimum when all cases in the node fall into a single category

Decision Trees – Approach: CART

Candidate split A: $\text{salary} \geq 50$

	branch	n	decline(0)	accept(1)	gini
0	$\text{salary} \geq 50$	9	5	4	?
1	$\text{salary} < 50$	7	7	0	?

Before	decline	accept
	12	4

Candidate split B: $\text{commute} \leq 1\text{h}$

	branch	n	decline(0)	accept(1)	gini
0	$\text{commute} \leq 1\text{h}$	13	9	4	?
1	$\text{commute} > 1\text{h}$	3	3	0	?

Decision Trees – Approach: CART

$$GINI(t) = 1 - \sum_j^{n_c} [p(j|t)]^2$$

$$\begin{aligned} \text{Gini (before)} &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(12/16)^2 + (4/16)^2] \\ &= 0.375 \end{aligned}$$

Candidate split A: salary ≥ 50

Before	decline	accept
	12	4

	branch	n	decline(0)	accept(1)	gini
0	salary ≥ 50	9	5	4	?
1	salary < 50	7	7	0	?

$$\begin{aligned} \text{Gini}(s_0) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(5/9)^2 + (4/9)^2] \\ &\sim 0.494 \end{aligned}$$

$$\begin{aligned} \text{Gini}(s_1) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(7/7)^2 + (0/7)^2] \\ &= 0 \end{aligned}$$

Candidate split B: commute $\leq 1h$

	branch	n	decline(0)	accept(1)	gini
0	commute $\leq 1h$	13	9	4	?
1	commute $> 1h$	3	3	0	?

$$\begin{aligned} \text{Gini}(c_0) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(9/13)^2 + (4/13)^2] \\ &\sim 0.426 \end{aligned}$$

$$\begin{aligned} \text{Gini}(c_1) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(3/3)^2 + (0/3)^2] \\ &= 1 \end{aligned}$$

Question: How to aggregate the branch-based values for the whole split?

Decision Trees – Approach: CART

- When a node p is split into k partitions (children), the quality of split is computed as,

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i) \quad (1)$$

where

- k is number of branches
- $GINI(i)$ is the i -th branch-based GINI
- n_i = number of records at child i ,
- n = number of records at node p .

Decision Trees – Approach: CART

$$GINI(t) = 1 - \sum_j^{n_c} [p(j|t)]^2$$

$$\begin{aligned} \text{Gini (before)} &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(12/16)^2 + (4/16)^2] \\ &= 0.375 \end{aligned}$$

Candidate split A: salary ≥ 50 = 0.375

Before	decline	accept
	12	4

	branch	n	decline(0)	accept(1)	gini
0	salary ≥ 50	9	5	4	?
1	salary < 50	7	7	0	?

$$\begin{aligned} \text{Gini}(s_0) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(5/9)^2 + (4/9)^2] \\ &\sim 0.494 \end{aligned}$$

$$\begin{aligned} \text{Gini}(s_1) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(7/7)^2 + (0/7)^2] \\ &= 0 \end{aligned}$$

GINI_split = 0.2777777777777778 Gain = 0.0972222222222221

Candidate split B: commute $\leq 1h$

	branch	n	decline(0)	accept(1)	gini
0	commute $\leq 1h$	13	9	4	?
1	commute $> 1h$	3	3	0	?

$$\begin{aligned} \text{Gini}(c_0) &= 1 - [p(d)^2 + p(a)^2] \\ &= 1 - [(9/13)^2 + (4/13)^2] \\ &\sim 0.426 \end{aligned}$$

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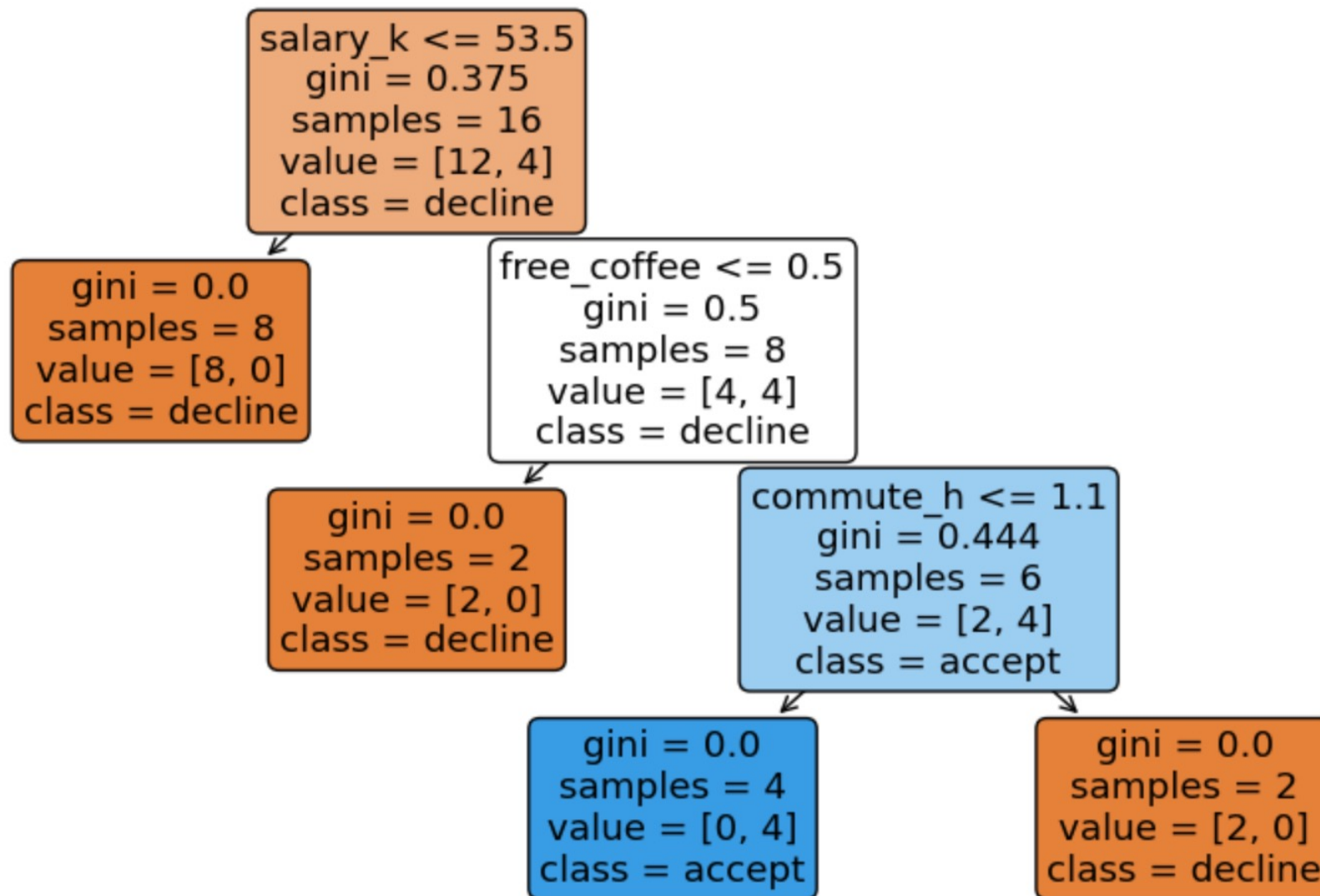
GINI_split = 0.3461538461538462 Gain = 0.028

Better split (smaller GINI_split): salary ≥ 50

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i) \quad (1)$$

Decision Trees – Approach: CART

CART tree learned from the offer dataset



Demo: Decision-trees_Offer_Case

Decision Trees – Approach: ID3

Similar to CART, Iterative Dichotomy 3 (ID3) minimises entropy

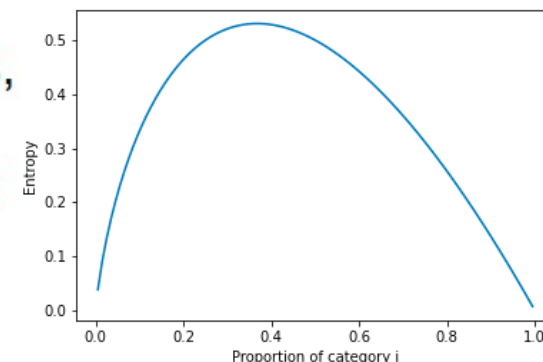
- Entropy at a given node t:

The smaller
the purer

$$Entropy(t) = - \sum_{j=1}^{n_c} p(j|t) \log_2 p(j|t) \quad (1)$$

(where: n_c is the number of classes, and $p(j | t)$ is the relative frequency of class j at node t).

- Measures **homogeneity** of a node.
 - Maximum ($\log_2 n_c$) when records are equally distributed among all classes implying the **least** information
 - Minimum (**0.0**) when all records belong to one class, implying the **most** information
- Entropy based computations are similar to the GINI index computations



Decision Trees – Approach: ID3

The higher
the better

- Information Gain:

$$GAIN_{split} = Entropy(p) - \left(\sum_{i=1}^k \frac{n_i}{n} Entropy(i) \right) \quad (1)$$

where $Entropy(p)$: entropy of Parent Node, $Entropy(i)$:
entropy of the i -th branch/partition, n is split into k partitions;
 n_i is number of records in partition i

- Measures **Reduction in Entropy** achieved because of the split. Choose the split that achieves most reduction (maximizes GAIN)
- Used in ID3 and C4.5

Decision Trees – Approach: ID3

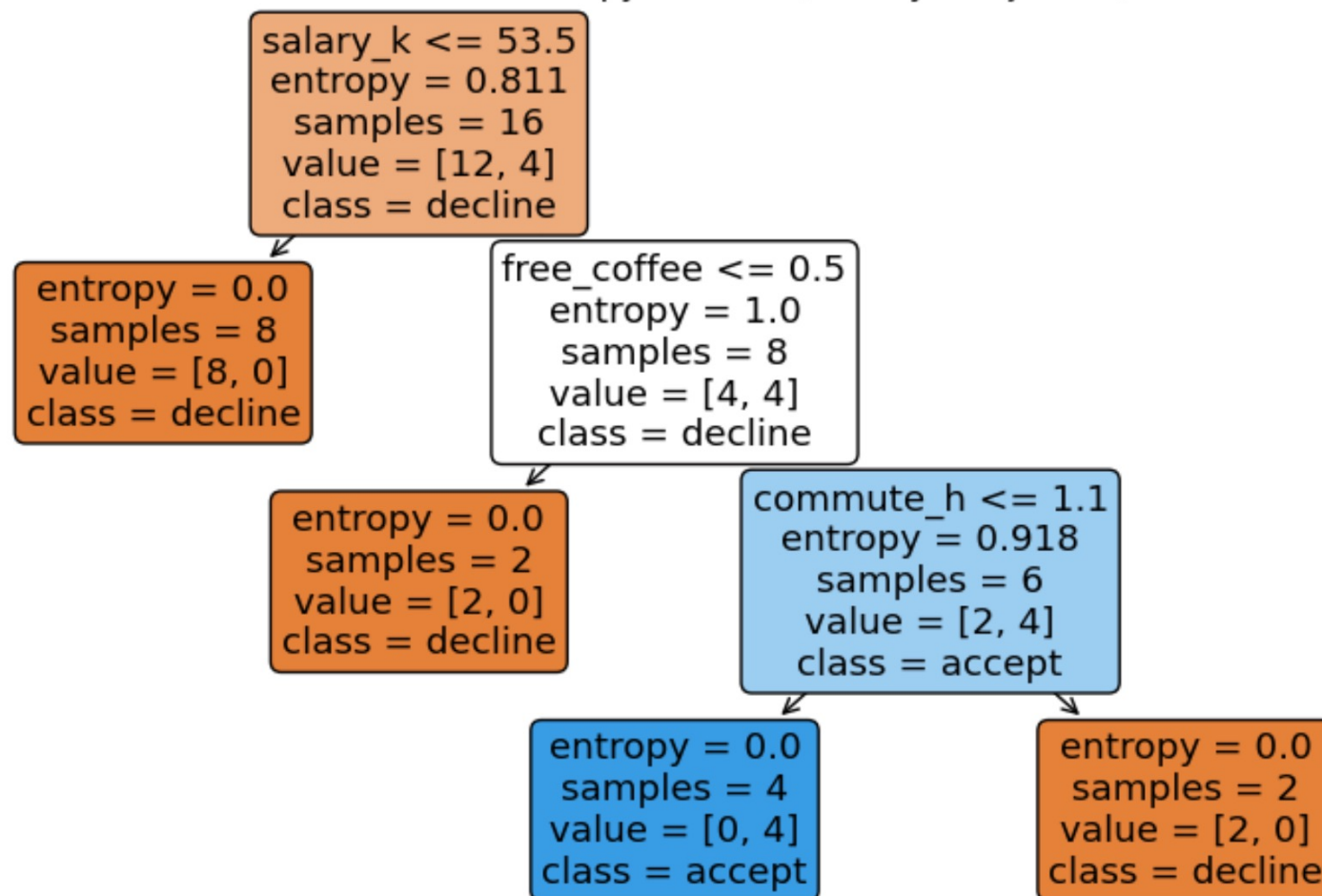
Entropy(parent) = 0.8112781244591328

File display gain (salary $\geq$ 50) = 0.2537978408001328

Entropy gain (commute $\leq$ 1h) = 0.0877536667807961

Better split (higher gain): salary $\geq$ 50

Tree learned with Entropy criterion (ID3-style objective)



Decision Trees – Overfitting Problem

We generate a larger **offer-themed** dataset with label noise and irrelevant features, then compare:

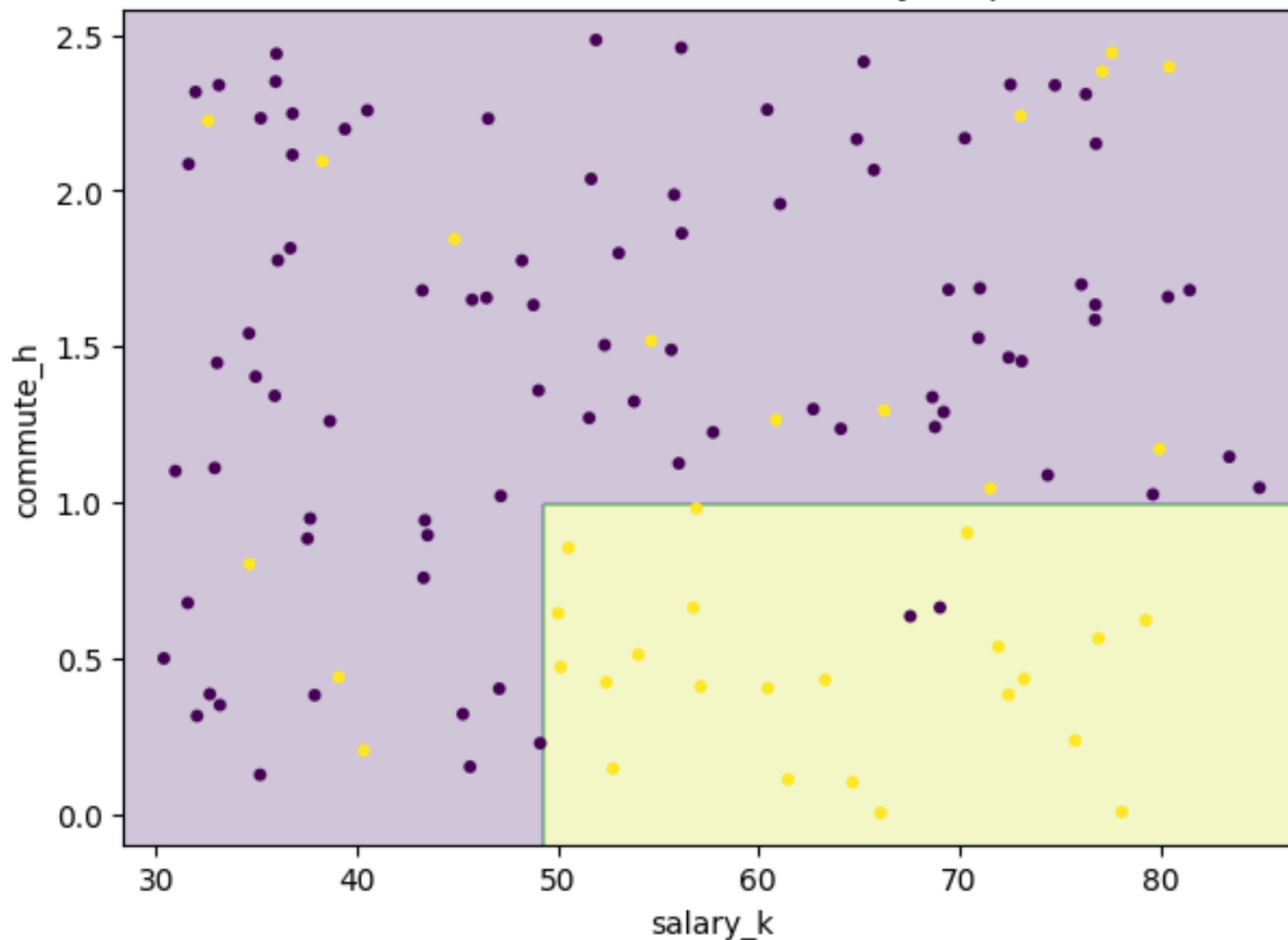
- a shallow tree (regularised)
- a deep tree (can overfit)

We also visualise a 2D boundary (salary vs commute) to make overfitting intuitive.

	salary_k	commute_h	free_coffee	bonus_k	team_size	office_plants	snack_score	accept
0	72.567583	1.846451	1	12.640980	12	13	0.446461	0
1	54.138314	0.792883	1	6.695842	29	19	-0.060092	1
2	77.222886	2.225049	1	14.294627	3	7	0.012579	0
3	68.355242	1.484576	0	9.761919	16	23	0.656819	0
4	35.179754	0.315066	0	1.738483	19	20	-0.727368	0

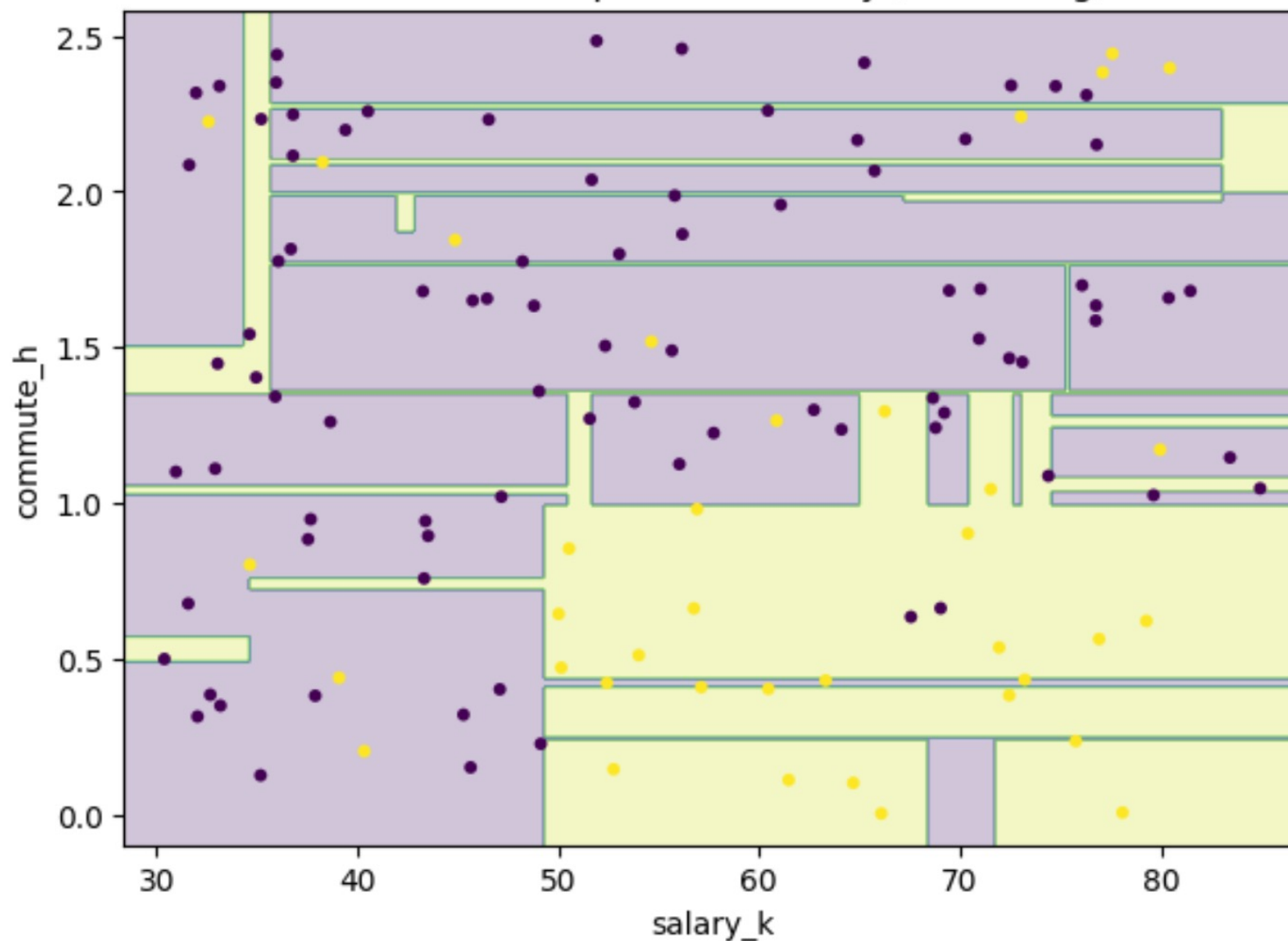
Decision Trees – Overfitting Problem

Offer case: shallow tree boundary (depth=2)



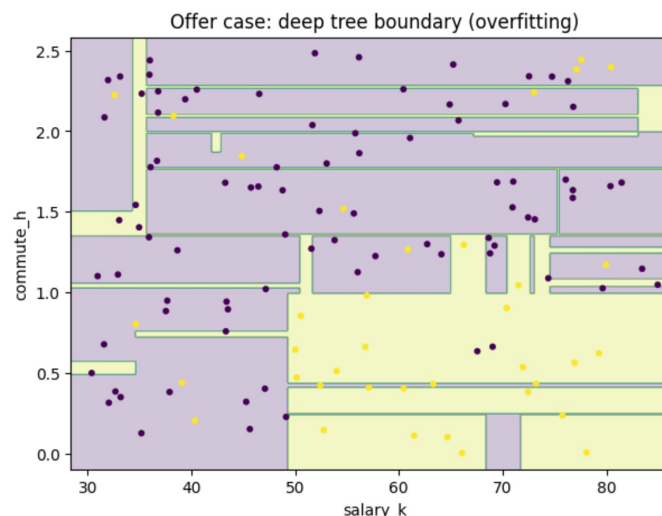
Decision Trees – Overfitting Problem

Offer case: deep tree boundary (overfitting)



Decision Trees – Pruning Approach

Overfitting is a serious problem with Decision Trees



.. Are these splits really required here?

The trees it creates are too complex.

One solution is **pruning**

This can be done in a variety of ways, including:

- ▶ Reduced Error Pruning
- ▶ Entropy Based Merging

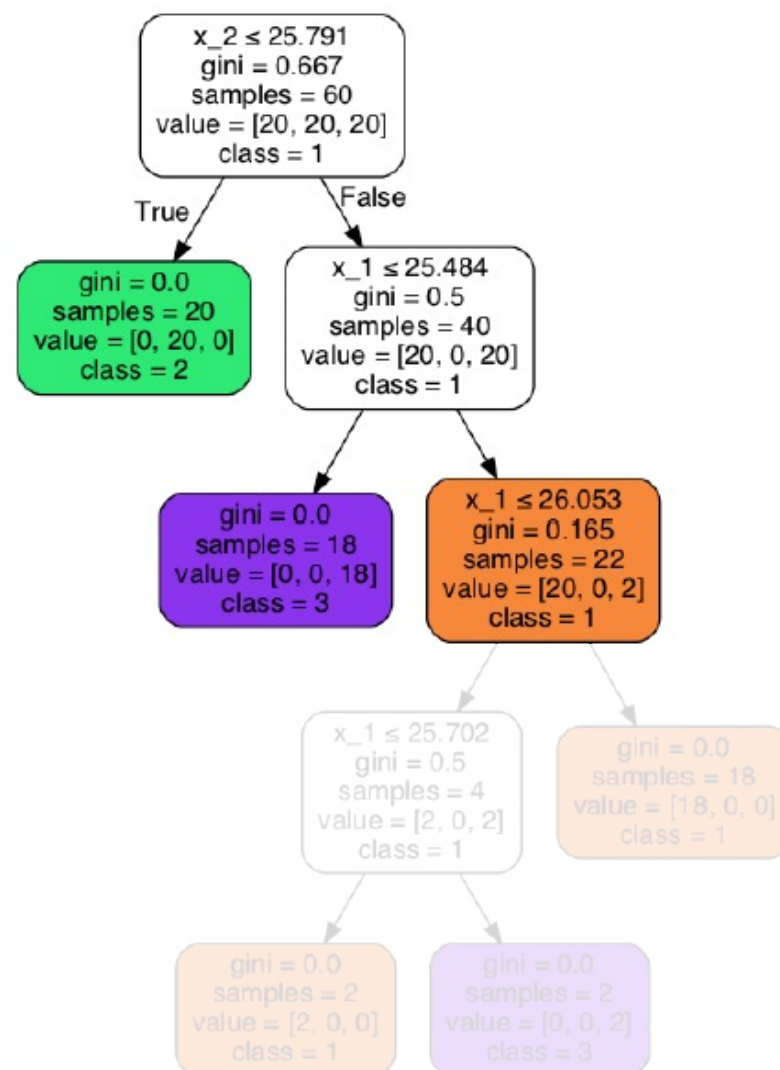
Decision Trees – Pruning Approach

Reduced Error Pruning

Growing a decision tree fully, then removing branches without reducing predictive accuracy, measured using a *validation set*.

- ▶ Start at leaf nodes
- ▶ Look up branches at last decision split
- ▶ replace with a leaf node predicting the majority class
- ▶ If validation set classification accuracy is not affected, then keep the change

This is a simple and fast algorithm that can simplify over complex decision trees



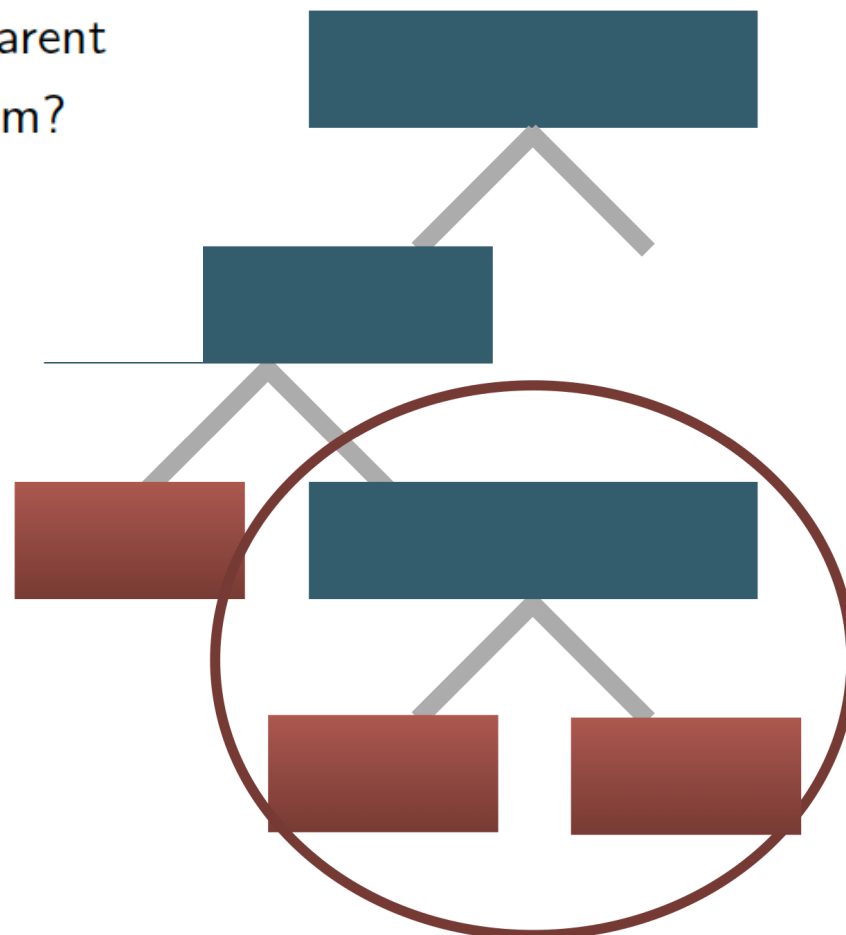
Decision Trees – Pruning Approach

Entropy Based Pruning

Grow a decision tree fully, then

- ▶ Chose a pair of leaf nodes with the same parent
- ▶ What is the entropy gain from merging them?
- ▶ If lower than a threshold, merge nodes

This doesn't require additional data



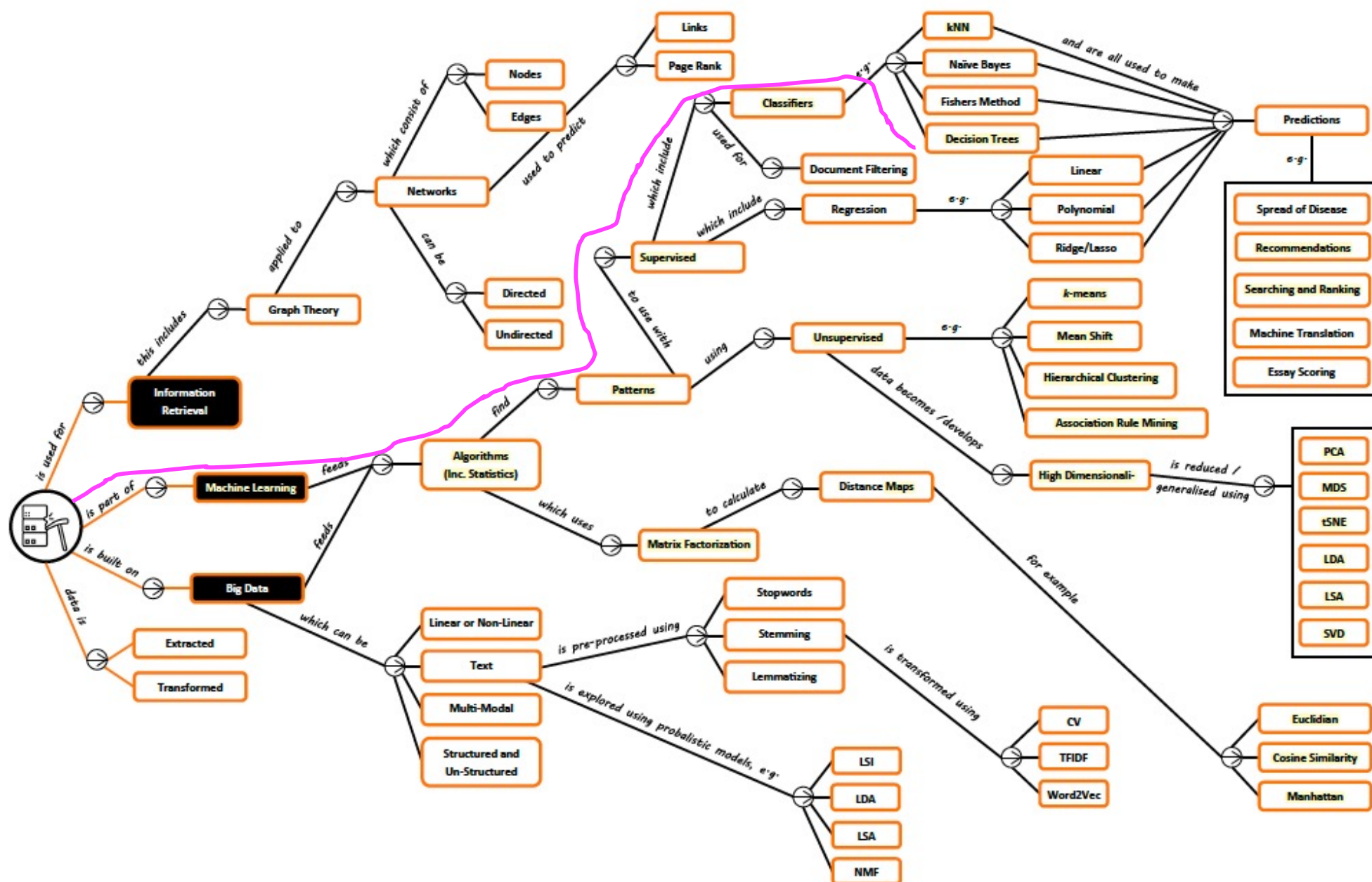
Decision Trees – Summary

	model	test_acc
2	Random forest	0.875000
1	Bagging	0.854167
4	Gradient boosting	0.854167
3	AdaBoost (stumps)	0.829167
0	Single tree	0.808333

- **CART** minimises impurity (e.g., Gini) to choose splits.
- **ID3** maximise information gain (entropy reduction).
- Trees can **overfit**; pruning reduces complexity.
- **Ensembles** (bagging, random forests, boosting) usually improve generalisation.
(at appendix, useful for coursework, not assessed in the exam.)

Side note: decision tree algorithms can also perform regression by partitioning the data into smaller, more homogeneous regions where the target variable has a similar numeric value. Unlike classification, which uses categories, regression trees aim to minimize variance within each split

Decision Trees – Roadmap



Decision Trees – Textbook

PART FOUR CLASSIFICATION

The classification task is to predict the label or class for a given unlabeled point. Formally, a classifier is a model or function M that predicts the class label $\hat{y}$ for a given input example $\mathbf{x}$, that is, $\hat{y} = M(\mathbf{x})$, where $\hat{y} \in \{c_1, c_2, \dots, c_k\}$ and each c_i is a class label (a categorical attribute value). Classification is a *supervised learning* approach,

- Data Mining and Machine Learning: Fundamental Concepts and Algorithms M. L. Zaki and W. Meira

4 Classification: Alternative Techniques

The previous chapter introduced the classification problem and presented a technique known as the decision tree classifier. Issues such as model overfitting and model evaluation were also discussed. This chapter presents alternative techniques for building classification models—from simple techniques such as rule-based and nearest neighbor classifiers to more sophisticated techniques such as artificial neural networks, deep learning, support vector machines, and ensemble methods. Other practical issues such as the class imbalance and multiclass problems are also discussed at the end of the chapter.

- Introduction to Data Mining, *P. Tan et al*

Decision Trees – Learning Outcomes

- **LO1:** Demonstrate an understanding of decision tree fundamentals, such as (exam)
 - Calculating impurity and constructing a decision tree using algorithms like ID3, C4.5, etc, given a dataset and distance metric
 - Addressing overfitting in decision trees like pruning
 - Understanding ensemble methods using decision trees like bagging, boosting, random forests
- **LO2: Implement** the learned algorithms using decision trees (course work)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*

Appendix – Ensemble methods

Bagging: Bootstrap aggregating

Uniformly sample initial dataset with replacement into m subsets

For example:

if data set has 5 samples, $(s_1, s_2, s_3, s_4, s_5)$

make a whole bunch of similar data sets:

▶ $(s_5, s_2, s_2, s_1, s_5)$

▶ $(s_4, s_2, s_1, s_3, s_3)$

▶ $(s_2, s_5, s_3, s_1, s_1)$

▶ $(s_3, s_1, s_2, s_4, s_4)$

Train a different decision tree on each set

To classify, apply each classifier and choose the correct one by majority vote

Appendix – Ensemble methods

Boosting - Kearns and Valiant (1988):

“Can a set of weak learners create a single strong learner?”

We make a weighted sum of very weak learners

- so long as they all learn different things then it works!

AdaBoost:

- ▶ Train a weak learner on one sampled data set
- ▶ See what it does well on
- ▶ Weight the difficult data samples more
- ▶ repeat

This makes a series of weak learners that have learned how to use given features to classify easy and hard samples more correctly

Appendix– Ensemble methods

Gradient boosting with trees - Friedman (1999):

Generalise Adaboost to Gradient boosting to handle any loss function

Shortcomings are where the residuals are larger

So fit a tree to the residuals:

▶ $x_1, y_q - f(x_1)$

▶ $x_2, y_q - f(x_1)$

▶ $x_3, y_q - f(x_1)$

▶ $\vdots$

▶ $x_n, y_q - f(x_n)$

more detail available at

http://www.chengli.io/tutorials/gradient_boosting.pdf

Appendix – Ensemble methods

Random Forests

Apply bagging

but when learning the tree for each subset, chose the split by searching over a random sample of the features

Reduces overfitting

[Demo: Decision-trees_Offer_Case](#)